

Kinabalu Basin Dossier v2

KINABALU BASIN — OFFSHORE NW SABAH, MALAYSIA

A Comprehensive Geological Dossier — v2.0 (October 2026)

Prepared for Arif Fazil · Revision of the July 2026 dossier v2 change basis: Lunt 2022 (BGSM 74) · Morley et al. 2023 (Geosphere) · Cottam et al. 2010 (JGS) · Bait 2003 (BGSM 47, verified) · Krebs 2011 (BGSM 57). Ages per GEOX event spine v0.1. Internal/partner well insights are deliberately excluded from this public document.

1 · What Is the Kinabalu Basin?

The Kinabalu Basin is the offshore sedimentary basin along the northwest Sabah margin — from the waters off Labuan Island north toward the Malaysia-Philippines maritime boundary. In the current institutional framework (Integrated Basin Study, Phase I), it **comprises three sub-basins: the Inboard Belt, the Outboard Belt, and the East Baram Delta**. It is one of only three hydrocarbon-producing basins in Malaysia (with the Malay and Sarawak basins); the remaining five (Langkasuka, Penyu, Layang-Layang, Sandakan, Tawau) are frontier. The basin spans the NW Sabah shelf (<100 m water) to the deepwater fold-thrust belt (>1,000 m); its northwestern limit is the Sabah Trough — a flexural foredeep over thinned Dangerous Grounds crust, not a simple trench.

Key numbers · Area ~25,000 km² explored · >10 km sediment in depocentres · Status: producing, oil and gas · Fields: Kinabalu (~500 MMstb oil-in-place, Bait 2003), Samarang, Kebabangan,

Sumandak, Malikai · Recent (2024–26, per EAGE abstracts): Zoisit Deep-1, Well A (deeper oil), Well B (deeper targets, multiple plays).

2 · Tectonic Evolution — nine phases, revised ages

Plate setting. Sabah sits at the junction of the Eurasian, Indo-Australian, Pacific and Philippine Sea plates, between the South China Sea (west), Sulu Sea (northeast) and Celebes Sea (east).

Phase 1 — Proto-South China Sea subduction (Eocene-Oligocene). SE-directed subduction beneath NW Borneo; deep-marine turbidites of the Rajang-Crocker Group (East/West Crocker, Trusmadi, Sapulut formations) accreted into the structural backbone of Sabah.

Phase 2 — Sarawak Orogeny (~40 Ma). Collision-driven uplift; West Crocker recycles Rajang detritus; Crocker-basin deep-water sedimentation ceases.

Phase 3 — Forearc deposition & mélange (Late Oligocene-Early Miocene). Labang and Kulapis formations with syn-depositional growth faults; Kuamut, Garinono and Ayer mélanges develop.

Phase 4 — Collision onset: the BMU (~24 Ma). The Dangerous Grounds continental block enters the subduction zone; uplift and erosion cut the **Base Miocene Unconformity (BMU, ~24 Ma** — Lunt 2022, BGSM 74; older “22–20 Ma” labelling revised). Deep-water conditions persist offshore above the BMU — the basin does *not* flip to shallow marine everywhere at this boundary.

Phase 5 — Stage III deep-marine phase & wedge growth (~24 → 13 Ma). (*New in v2 — previously missing.*) Between BMU and DRU the margin hosts a thick deep-marine succession (Setap Shale / Temburong-equivalent), imbricated into the accretionary wedge. In northern Sabah, the North Sabah-Pagasa Wedge (NSPW) grows by underthrusting of Dangerous Grounds crust — mini-basins, mobile shale, overpressure (Morley et al. 2023). *Scope: NSPW phases are documented for northern Sabah only; southern-Sabah deformation mostly post-dates the SRU.*

Phase 6 — The DRU (~13–12 Ma) — a composite surface, and NOT the Sarawak MMU. The **Deep Regional Unconformity is now placed at roughly 13–12 Ma** (Lunt 2022, BGSM 74: bracket 13.77–11.7 Ma; Morley et al. 2023: 13.5–12.5 Ma). It is **several million years younger than the Sarawak MMU (~16 Ma)** — the two names became mixed in the literature, and the legacy “15.5 Ma” age traces to a rounding artefact propagated

from early schematics. The older reading ties the DRU to the Middle Miocene Cagayan Arc-Palawan collision at ~15.5 Ma (Balaguru & Hall 2009) — retained as a documented minority interpretation. Critically, Morley et al. (2023) show the DRU, UIU and SRU are “**mixtures of conformable regions together with local unconformities rather than being simple widespread unconformities**” — one name, several surfaces. This is the core justification for the IBS single-NN-standard recalibration.

Phase 7 — Stage IV deltas, mud canopy & the UIU/SRU (Late Miocene). Stage IV shelf-deltaic clastics prograde northwestward (main Kinabalu Field reservoirs). In northern Sabah the mud canopy coalesces **at the UIU (~10.5 Ma) in a short burst**, and Stage IVC turbidites route around canopy relief; the **SRU (~8.5-9 Ma)** caps the canopy-basin fill (Morley et al. 2023).

Phase 8 — Kinabalu granite (7.85-7.22 Ma) — post-SRU. The Kinabalu granodiorite is a **sheeted laccolith of dyke-fed sheets younging downward, emplaced in <800 ka** (U-Pb zircon; Cottam et al. 2010, JGS 167) — *younger than the SRU*, driven by Sulu Arc rollback, and **not** related to the Sabah orogeny. Today it stands as Southeast Asia’s highest peak (4,095 m).

Phase 9 — Late Pliocene-Recent transpression (unnamed) & gravitational collapse. NW-SE strike-slip/transpressional deformation — expressed in the “circular basin” outcrop pattern and in active faults (2015 Ranau Mw 6.0 earthquake; GPS shows both compression and extension). The <10 Ma deformation engine is gravitational collapse of the uplifted orogen plus shale-detachment gravity tectonics. (*v2: the informal “Meliau Orogeny” label is dropped as an unverified coinage.*)

3 · Stratigraphy Simplified

Youngest → oldest: **Stage IVG-IVA** (Late Miocene shelf-deltaic cycles — IVC coastal progradation and growth-fault sands; IVD stacked coarsening-upward shoals; IVE transgression with shore-parallel sands; IVF widespread transgressive sheets; IVG renewed shallowing) · **SRU ~8.5-9 Ma** · Stage IVB-IVC (deepwater Kinabalu-belt turbidites, canopy-era) · **UIU ~10.5 Ma · Stage III (Setap/Temburong-equivalent deep-marine, ~24-13 Ma)** · **DRU ~13-12 Ma (composite)** · **BMU ~24 Ma** · Rajang-Crocker Group and mélange (pre-BMU).

4 • Kinabalu Field (all figures Bait 2003, BGSM 47 — verified)

Discovered 1989 by KN-1 (1,043 ft net oil sand, 113 ft net gas, 310 ft undifferentiated HC sand); 55 km WNW of Labuan in ~54 m water. ~**500 MMstb oil-in-place**; first oil December 1997; peak 48,000 bopd; ~50 MMstb produced by 2003. Three accumulations: **Kinabalu Main** (oil in 30+ sands: F, J, K, L, M, O; normally pressured), **Kinabalu Deep** (condensate-rich gas, ~250 ft oil rim; mild overpressure; separated by 1,700 ft of water-bearing section), **Kinabalu East** (gas with thin oil rims). Trap: dip-closure against the SW-NE Kinabalu Growth Fault (~8° WNW dip; NE spill point). L-sand lithofacies: poorly stratified (ϕ 23%, k630 mD — best), laminated (ϕ 21%, k126 mD), bioturbated (ϕ 16%, k11 mD). Water injection: B/C-aquifer dump-flood via two horizontal wells (natural ~1,200 bpd; ESP to 20,000 bpd), lifting L-reservoir recovery toward 43% (+~16 MMstb).

5 • Other Fields & Recent Discoveries

Samarang (adjacent; oil & gas; Stage IV; shared Samarang complex→Labuan OT) • **Kebabangan** (gas; receives Malikai tieback; exports to SOGT) • **Sumandak** (2001; Stage IVC) • **Malikai** (deepwater oil via KGB platform). **2024-26 (EAGE-abstract level)**: Zoisit Deep-1; **Well A** — oil in deeper intervals beneath known gas, shifting perception from gas-dominant to oil-prone; **Well B** — reprocessed-seismic deeper targets across varied play types.

6 • Petroleum System

Source — never directly penetrated; Type III/II-III, deltaic; Ro 0.7-0.8%; probable depth >3,200 m; generation Late Miocene–Recent. **Reservoir** — primary: **post-DRU** Stage IV clastics (ϕ 20–30%, $k \leq 630$ mD); secondary: **pre-BMU/pre-DRU** Oligocene syn-rift clastics and carbonates (public record: Tepat-1 gas + oil rim, Nido-equivalent buildup, Layang-Layang basin — TGS 2026). **Seal** — intra-formational Stage IV shales; IVF/IVG transgressions regional; DRU + shales basin-scale. **Trap** — fault-dependent dip closures, syn-kinematic anticlines (fold-thrust belt); stratigraphic traps underexplored. **Timing risk** — charge vs Late Miocene–Pliocene transpression: seal breach/remigration is a present-tense process (Sandakan analogue: 5/19 wells).

7 · Regional Cross-Section — NW-SE Transect (~600 km)

Dangerous Grounds — extended continental crust, block-faulted, thin cover; the colliding block (arrival ~24 Ma). **Layang-Layang basin** — Oligocene syn-rift half-grabens with carbonate buildups on highs; revitalised by the Tepat-1 discovery (public record). **Sabah Trough** — flexural foredeep over underthrust Dangerous Grounds crust; conical seamounts, variously read as volcanoes or mud volcanoes (Hutchison 2010). **Kinabalu Basin** — inboard fold-thrust belt; post-DRU Stage IV plays; Kinabalu Growth Fault (active ~8.5–6.7 Ma); toe-thrusts at the basinward edge. **Crocker Range** — uplifted accretionary prism, intruded by the Kinabalu sheeted laccolith (7.85–7.22 Ma). **Central Sabah** — Neogene “circular basins” (Tanjong/Kapilit delta remnants on mélange; Pliocene-Recent transpression). **Sandakan basin** → **Sulu Sea** — Sebahat progradation; five sub-commercial gas discoveries.

8 · References (v2 additions marked ★)

Balaguru & Hall (2009) S&D #30084 · Balaguru, Nichols & Hall (2003) BGSM 46/47 · Bait (2003) BGSM 47, 165–179 · Madon & Jong (2022) BGSM 74, 135–160 · PETRONAS MPM (2025) · Choi/TGS (2026) EAGE Workshop · Ab Ghani et al. (2026) EAGE Workshop · Madon et al. (1999) PETRONAS · Tongkul (1991) JSEES 6(3/4) · Hutchison (1996) GSL SP 106 · Hall (1996) GSL SP 106 · ★ **Lunt, P. (2022) BGSM 74, 39–55 — unconformity ages (BMU ~24 Ma; DRU ~13–12 Ma bracket 13.77–11.7) · ★ Lunt & Madon (2017) BGSM 64, 101–122 · ★ Morley, C.K. et al. (2023) Geosphere 19(1) — NSPW mud canopy; DRU 13.5–12.5; composite-surface doctrine; seamounts · ★ Cottam, M. et al. (2010) JGS 167, 49–60 — Kinabalu granite U-Pb 7.85–7.22 Ma · ★ Krebs, W.N. (2011) BGSM 57, 39–46 — chronosequence framework (version of record) · ★ Hutchison, C.S. (2010) Marine Geology — NW Borneo Trough.**

KINABALU BASIN DOSSIER v2.0 · October 2026 · public sources only — internal/partner well data deliberately excluded · cross-section: arif-fazil.com/earth/kinabalu-cross-section.html · ages: GEOX event spine v0.1